

HTML Tags & HTML Controls

Lab Notes — Formatting, Structural, and Embedding Tags with Runnable Code

This document explains each tag/control from the lab, with the exact HTML code, what it does, and where it's typically used. A separate, runnable .html file has been created for every numbered task (1_text_formatting.html through 8_list_controls.html) — open any of them directly in a browser. Everything below is editable in Word.

Task	Covers	File
1 (A)	Text formatting: b, i, u, strong, em	1_text_formatting.html
2 (B)	sup / sub for math & chemical expressions	2_math_chemical_expressions.html
3 (C)	pre and code tags for preformatted text/code	3_pre_code_demo.html
4 (A)	Headings h1-h6 and paragraphs	4_headings_paragraphs.html
5 (A)	address, b, u, i, blockquote, marquee, link, meta	5_address_bui_blockquote_marquee_link_meta.html
6 (B)	Anchor tag and image control	6_anchor_image.html
7 (B)	Iframe tag	7_iframe_demo.html
8 (C)	List controls (ul, ol, nested, dl)	8_list_controls.html

1. Text Formatting Tags — b, i, u, strong, em

Purpose: These tags change how a piece of text is displayed or emphasised, without creating a new block of content.

Tag	Effect	Meaning conveyed?
<code></code>	Bold text	Visual only — no semantic weight
<code><i></code>	Italic text	Visual only — no semantic weight
<code><u></code>	Underlined text	Visual only — no semantic weight
<code></code>	Bold text	Marks content as important (read differently by screen readers)
<code></code>	Italic text	Marks content as emphasised/stressed

```
<p><b>Bold text</b></p>
<p><i>Italic text</i></p>
<p><u>Underlined text</u></p>
<p><strong>Strongly important text</strong></p>
<p><em>Emphasised text</em></p>
```

Recap / Extra: Prefer `<strong>` and `<em>` over `<b>` and `<i>` in modern HTML — they carry meaning for accessibility tools and search engines, not just appearance.

2. sup and sub — Mathematical & Chemical Expressions

Purpose: <sup> raises text above the baseline (superscript); <sub> lowers text below the baseline (subscript). Used for exponents, chemical formulas, and footnotes.

```
<p>Water: H<sub>2</sub>O</p>  
<p>Carbon dioxide: CO<sub>2</sub></p>  
<p>Square of x: x<sup>2</sup></p>  
<p>Einstein's equation: E = mc<sup>2</sup></p>
```

Example: Combined use: CH₄ + 2O₂ -> CO₂ + 2H₂O (methane combustion) needs <sub> for every atom count; Avogadro's number 6.022 x 10²³ needs <sup> for the exponent.

3. pre and code — Preformatted Text & Code Snippets

Purpose: Normal HTML collapses extra spaces and line breaks into one space. `<pre>` preserves whitespace and line breaks exactly as typed. `<code>` marks text as computer code (usually monospace font). The two are commonly combined: `<pre><code>...</code></pre>`.

```
<p>Use <code>document.getElementById()</code> to select an element.</p>

<pre><code>function greet(name) {
  console.log("Hello, " + name + "!");
}

greet("World");</code></pre>
```

Recap / Extra: Without `<pre>`, the indentation and line breaks in the function above would all collapse onto one line — this is why `<pre>` is essential for showing code or tabular plain text.

3.1 `<pre>` vs `<code>` — key differences

These two are often used together, but they solve different problems: `<code>` is a semantic, inline tag that marks text as computer code and renders it in monospace — it does not preserve whitespace or line breaks and does not create a block-level box. `<pre>` is a block-level tag that preserves preformatted text exactly as written (spaces, tabs, and line breaks all kept intact), and also renders in monospace by default.

Feature	<code><pre></code>	<code><code></code>
Type	Block-level	Inline
Preserves whitespace/line breaks	Yes	No
Purpose	Show formatted blocks of text (poetry, ASCII art, multi-line code)	Mark a short code snippet within a sentence
Common combo	<code><pre><code>...</code></pre></code> used together for multi-line code blocks	—

Recap / Extra: Best practice for a code block is to nest them: `<pre><code>...</code></pre>` — `<pre>` handles the formatting, `<code>` handles the semantic meaning.

4. Headings (h1-h6) and Paragraphs

Purpose: Headings define the outline/hierarchy of a page's content, from most important (<h1>) to least (<h6>). <p> wraps a block of regular paragraph text.

```
<h1>Page Title</h1>
<h2>Major Section</h2>
<h3>Sub-section</h3>
<h4>Deeper heading</h4>
<h5>Even deeper</h5>
<h6>Smallest heading</h6>
<p>Regular paragraph text goes here.</p>
```

Recap / Extra: Use only one <h1> per page (the main title) and don't skip levels (e.g. h2 straight to h5) — screen readers and SEO tools rely on a logical heading order, not just font size.

5. Address, B, U, I, Blockquote, Marquee, Link, Meta

Tag	Purpose	Notes
<address>	Marks contact information for the page/author	Usually rendered in italics by default
	Bold text	Visual only
<u>	Underlined text	Visual only
<i>	Italic text	Visual only
<blockquote>	Marks a long quotation from another source	Browsers usually indent it
<marquee>	Scrolling text (old, non-standard)	Deprecated — avoid in real projects, use CSS animation
<link>	Connects an external resource (e.g. CSS file) — used in <head>	Has no closing tag
<meta>	Page metadata: charset, description, viewport, etc.	Not visible on the page itself

```
<head>
  <meta charset="UTF-8">
  <meta name="description" content="Demo page">
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <address>Harsh Patel</address>
  <blockquote>The Web is more social than technical. - Tim Berners-Lee</blockquote>
  <marquee>Scrolling announcement text</marquee>
</body>
```

5.1 <marquee> — behavior attribute alternatives

Heads-up: <marquee> is a deprecated, non-standard HTML tag. Browsers still render it for legacy reasons, but it shouldn't be used in modern projects — CSS animations are the recommended replacement. That said, for a practical/demo exercise, the behavior attribute accepts these values instead of "scroll":

Value	Effect
scroll (default)	Text scrolls in from one side and off the other, then repeats
slide	Text slides in from one side and stops when it reaches the edge (doesn't loop out)
alternate	Text bounces back and forth between the two edges

```
<marquee behavior="alternate" direction="left">
  Welcome to HTML Practical Demonstration
</marquee>
```

6. Anchor Tag and Image Control

6.1 Anchor tag <a>

Purpose: Creates a hyperlink to another page, a section on the same page, an email address, or a downloadable file.

Attribute	Meaning
href	The destination URL, anchor (#id), mailto:, or file path
target="_blank"	Opens the link in a new browser tab
title	Tooltip text shown on hover
download	Forces the browser to download the file instead of navigating to it

```
<a href="https://www.wikipedia.org" target="_blank" title="Visit
Wikipedia">Wikipedia</a>
<a href="#section2">Jump to Section 2</a>
<a href="mailto:contact@example.com">Send an email</a>
<a href="notes.pdf" download>Download Notes</a>
```

6.2 Image control

Purpose: Embeds an image on the page. Unlike most tags, is a self-closing (void) element with no separate closing tag.

Attribute	Meaning
src	Path or URL of the image file
alt	Alternate text shown if the image fails to load; also read by screen readers
width / height	Sets the display size of the image

```


<!-- Image used as a clickable link -->
<a href="https://www.wikipedia.org">
  
</a>
```

7. Iframe Tag

Purpose: Embeds another HTML page inside the current page, in its own independent rendering area.

Attribute	Meaning
src	URL of the page being embedded
title	Accessible name for the frame — required for screen readers
width / height	Size of the embedded frame
sandbox	Restricts what the embedded page can do, for security

```
<iframe
  src="https://en.wikipedia.org/wiki/HTML"
  title="Wikipedia HTML article"
  width="600" height="350"
  sandbox="allow-scripts allow-same-origin">
</iframe>
```

Example: Common real-world uses: embedding a Google Map on a contact page, embedding a YouTube video, or embedding a third-party payment widget.

8. List Controls

Tag	Type	Use case
<code> / </code>	Unordered (bulleted) list	Items with no particular order — e.g. a feature list
<code><ul type="disc"></code> default	● Item 1	
<code><ul type="circle"></code> <code><ul type="square"></code>	○ Item 1 ■ Item 1	
<code> / </code>	Ordered (numbered) list	Sequential steps — e.g. instructions
<code><ol type="1"></code> default	1. Apple	
<code><ol type="A"></code> <code><ol type="a"></code> <code><ol type="i"></code> <code><ol type="I"></code>	A. Apple a. Apple i. Apple I. Apple	
<code><dl> / <dt> / <dd></code>	Description list	Term-definition pairs — e.g. a glossary

```
<ul>
  <li>HTML</li>
  <li>CSS</li>
</ul>

<ol>
  <li>Plan the page</li>
  <li>Write the HTML</li>
</ol>

<dl>
  <dt>HTML</dt>
  <dd>Defines page structure.</dd>
</dl>
```

Recap / Extra: Lists can be nested — placing a full `<ul>` or `<ol>` inside an `<li>` — to represent sub-items, exactly as shown in `8_list_controls.html`.